EXERCISE 7.8

(1) $6, 6$ (2) $2\sqrt{5}, 2\sqrt{5}$ (3) 50 (4) 100m^2 (5) 9cm , 6cm (6) 1200m

(7) $10\sqrt{2}, 10\sqrt{2}$ (9) $\sqrt{2}r, \frac{r}{\sqrt{2}}$ (10) 6cm , 6cm , 3cm (11) $32\pi, 0$

EXERCISE 7.9

(1) (i) $x=-1, x=1, y=1$ (ii) $x=-1, y=x-1,$ (iii) $y=-3, y=3$

(iv) $y=x-9, x=-3$

EXERCISE 7.10

1	2	3	4	5	6	7	8	9	10
(1)	(2)	(2)	(2)	(1)	(2)	(3)	(4)	(3)	(1)
11	12	13	14	15	16	17	18	19	20
(3)	(4)	(3)	(4)	(2)	(3)	(3)	(1)	(4)	(3)

Chapter 8

Exercise 8.1

1. (i) 3.0074 2. (i) 24.73 (ii) 1.9688 (iii) 2.963
3. (i) $7x-4$ (ii) $\frac{9-4x}{5}$ (iii) $\frac{x+1}{4}$
4. (i)0.0225π cm^2, (ii) 0.006 cm^2 (iii) 0.6%
5. (i) Volume decreases by $80\pi\ cm^3$ (ii) Surface area decreases by $16\pi\ cm^2$ 6. 1%

Exercise 8.2

1. (i) $\frac{2(1-2x)^2(8x-7)}{(3-4x)^2}dx$ (ii) $\frac{4}{3}\frac{\cos 2x}{(3+\sin 2x)^{\frac{1}{3}}}dx$ (iii) $e^{x^2-5x+7}[(2x-5)\cos(x^2-1)-2x\sin(x^2-1)]dx$
2. (i) 0.7 (ii) 0.18 3. (i) $\Delta f = 3.125$, $df = 2.0$ (ii) $\Delta f = 0.11$, $df = 0.1$
4. 3.0013029 5. (i) $\frac{6}{\pi}$ cm (ii) $\frac{40}{\pi}\%$ 6. $30\pi\ mm^3$ 7. $0.4\pi\ mm^2$ 8. 8000
9. (i) $\simeq 3$ words (ii) $\simeq 1$ word 10. 5.25π , 4.76% 11. 60 cm^3 , 61.2 cm^3

Exercise 8.3

1. $\frac{1}{8}$ 2. 1 4. cos(1)

Exercise 8.4

1. (i) 27, −14 (ii) 11, −4 (iii) 2, 0, 4 (iv) $e^2\left((\log 2)-1\right), e^2\left(1+\log 8\right)$
3. $\frac{x^2-y^2}{x^2y}, \frac{y^2-x^2}{y^2x}+3z^2$, 6yz 4. $\frac{3\left(x^2+y^2+z^2\right)}{\left(x^3+y^3+z^3\right)}$
5. (i) e^y+6x , $6y$, xe^y , e^y+6x

(ii) $\frac{-15}{(5x+3y)^2}$, $\frac{-25}{(5x+3y)^2}$, $\frac{-9}{(5x+3y)^2}$, $\frac{-15}{(5x+3y)^2}$ (iii) 3, $2-25\cos 5x$, 0, 3

10. (i) $72x+84y+0.04xy-0.05x^2-0.05y^2-2000$ (ii) 24, −48 , Keeping y constant and increasing x increases profit.

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